

CHAPTER 11

TORSION OF SHAFTS

The moment of a force applied to a shaft which tends to twist or turn is termed the twisting moment, turning moment, or torque. Its magnitude is the product of the applied force and its effective leverage (like any other moment of force).

TORSION is the state of being twisted, therefore a shaft transmitting torque is said to be in torsion.

When a shaft is twisted, each lamination of cross-section (thin circular discs of the shaft) rotates very slightly relative to the next lamination, thus the shaft suffers shear strain because the particles which constitute the material tend to slide over each other, and shear stress is induced.

THE FUNDAMENTAL TORSION EQUATION follows a similar pattern and expresses similar relationships as the fundamental bending equation explained in the last chapter.

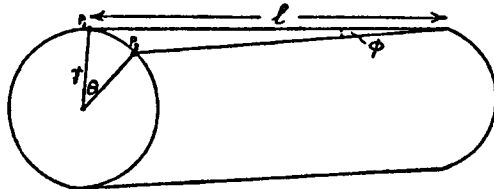


Fig. 194

Consider a shaft of length l twisted as shown in Fig. 194 such that as a result of the strain, a point on the circumference shifts from P_1 to P_2 . The amount of distortion is the arc length P_1 to P_2 and this depends upon the length l under twist. As shear strain is expressed by the angle of distortion (see Chapter 9), in this case angle ϕ , then,

$$\text{Shear strain at outer fibres} = \frac{P_1P_2}{l}$$

Let θ = angle of twist at shaft face

and r = radius from centre to outer fibres

then arc P_1P_2 = angle in radians $\times$ radius = θr

therefore,

$$\text{Shear strain at outer fibres} = \frac{\theta r}{l}$$

Provided the material is strained within the elastic limit, shear stress and shear strain are proportional to each other, and therefore shear stress divided by shear strain is a constant for any given material. This constant is termed the modulus of rigidity (G).

$$G = \text{shear stress} \div \text{shear strain}$$

$$= \tau \div \frac{\theta r}{l}$$

$$= \frac{\tau l}{\theta r}$$

Thus the shear stress at the outer fibres is expressed by:

$$\tau = \frac{G\theta r}{l} \quad \dots \dots \dots (i)$$

For any given section, θ and l are constant as well as G , the stress is therefore proportional to the radial distance from the centre. At the centre of the shaft the stress is nil, increasing uniformly to a maximum at the outer skin.

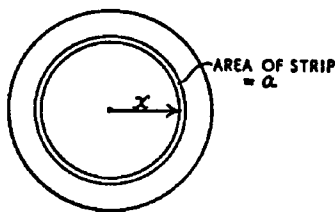


Fig. 195

Consider now a thin ring of the material of area a at the radius x from the centre (Fig. 195).

$$\text{Stress on this ring} = \frac{G\theta x}{l}$$

$$\text{Total load} = \text{stress} \times \text{area}$$

$$\therefore \text{Load on this ring} = \frac{G\theta x}{l} \times a$$

This load or force in the material exists as a resistance to support the externally applied twisting moment. The moment of resistance of the load carried by this ring of area is the product of the load and its leverage from the centre.

$$\begin{aligned}\text{Moment of resistance of ring} &= \frac{G\theta xa}{l} \times x \\ &= \frac{G\theta ax^2}{l}\end{aligned}$$

If the whole area of cross-section is considered to be made up of such thin rings a_1, a_2, a_3 etc., then the total moment of resistance offered by the whole area will be:

$$\frac{G\theta a_1 x_1^2}{l} + \frac{G\theta a_2 x_2^2}{l} + \frac{G\theta a_3 x_3^2}{l} + \text{etc.}$$

As G , θ and l are common to all rings across the section, this can be written:

$$\frac{G\theta}{l} \left\{ a_1 x_1^2 + a_2 x_2^2 + a_3 x_3^2 + \text{etc.} \right\}$$

$$\text{Total moment of resistance} = \frac{G\theta}{l} \Sigma ax^2$$

The summation of the products of elements of area and the square of their distances from a given point is the second moment of the whole area about that point. This second moment is taken about the polar centre of the shaft, it is therefore termed the polar second moment and can be represented by I_p or more usually by J .

Also, the total internal moment of resistance in the shaft must be equal to the externally applied twisting moment which is represented by T , therefore:

$$T = \frac{G\theta}{l} \times J$$

$$\text{or, } \frac{T}{J} = \frac{G\theta}{l} \quad \dots \dots \dots (ii)$$

Equation (i) can be written in a similar manner,

$$\frac{\tau}{r} = \frac{G\theta}{l}$$

and by combining (i) and (ii) gives the **fundamental torsion equation**,

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{l}$$

- T = twisting moment
 J = polar 2nd moment of area
 τ = shear stress at outer fibres
 r = radius to outer fibres
 G = modulus of rigidity
 θ = angle of twist
 l = length of shaft under twist

J values are derived by integral calculus (see Chapter 7).

For a solid circular section,

$$J = \frac{\pi}{32} D^4 \quad \text{or} \quad \frac{\pi}{2} R^4$$

For a hollow circular section,

$$J = \frac{\pi}{32} \left\{ D^4 - d^4 \right\} \quad \text{or} \quad \frac{\pi}{2} \left\{ R^4 - r^4 \right\}$$

Example. When transmitting a certain power an engine shaft 360 mm diameter is twisted 1° over a length of 4.5 m. If the modulus of rigidity of the shaft material is 103.5 GN/m^2 , find (i) the torque transmitted, (ii) the stress in the shaft.

$$\frac{T}{J} = \frac{G\theta}{l} \quad T = \frac{JG\theta}{l}$$

$$\begin{aligned}
 T &= \frac{\pi \times 0.36^4 \times 103.5 \times 10^9 \times 1}{32 \times 4.5 \times 57.3} \\
 &= 6.619 \times 10^5 \text{ N m} = 661.9 \text{ kN m} \quad \text{Ans. (i)}
 \end{aligned}$$

$$\frac{\tau}{r} = \frac{G\theta}{l} \quad \therefore \tau = \frac{rG\theta}{l}$$

$$\begin{aligned}
 \tau &= \frac{0.18 \times 103.5 \times 10^9 \times 1}{4.5 \times 57.3} \\
 &= 7.224 \times 10^7 \text{ N/m}^2 \\
 &= 72.24 \text{ MN/m}^2 \quad \text{Ans. (ii)}
 \end{aligned}$$

RELATION BETWEEN TORQUE AND STRESS

$$\frac{T}{J} = \frac{\tau}{r} \quad T = \frac{J\tau}{r}$$

For a solid round shaft,

$$J = \frac{\pi}{32} D^4 \quad \text{and} \quad r = \frac{1}{2} D$$

$$T = \frac{\pi \times D^4 \times 2 \times \tau}{32 \times D}$$

$$T = \frac{\pi D^3 \tau}{16}$$

For a hollow round shaft,

$$J = \frac{\pi}{32} (D^4 - d^4) \quad \text{and } r = \frac{1}{2}D$$

$$T = \frac{\pi (D^4 - d^4) \times 2 \times \tau}{32 \times D}$$

$$T = \frac{\pi (D^4 - d^4) \tau}{16D}$$

The stress in a shaft therefore depends upon the torque transmitted, the stress allowed depends upon the strength of the material, therefore the strength of shafts can be compared by the torque that can be transmitted as expressed in the given formulae.

Example. Compare the strengths and weights of two shafts of similar material, one is solid and the other is hollow of outside diameter equal to the diameter of the solid shaft with a bore of half that diameter.

STRENGTH RATIO:

Strength of solid shaft : Strength of hollow shaft

$$\frac{\pi D^3 \tau}{16} : \frac{\pi (D^4 - d^4) \tau}{16D}$$

π and 16 are common to both and therefore cancel. As the shafts are of similar material the stress allowed in each case is the same and also cancels, leaving:

$$D^3 : \frac{D^4 - d^4}{D}$$

$$D^4 : D^4 - d^4$$

Substituting $d = \frac{1}{2}D$,

$$D^4 : D^4 - (\frac{1}{2}D)^4$$

$$D^4 : D^4 - \frac{1}{16}D^4$$

$$D^4 : \frac{15}{16}D^4$$

Dividing both by D^4 ,

$$1 : \frac{15}{16} \quad \dots \dots \dots \quad (i)$$

To compare their weights,

Let l = length of each shaft (each of same length)

w = specific weight of material (same for each shaft)

Weight = area of section $\times$ length $\times$ spec. wt.

spec. wt. = $9.81 \times$ density

WEIGHT RATIO:

Weight of solid shaft : Weight of hollow shaft

$$0.7854 \times D^2 \times l \times w : 0.7854 (D^2 - d^2) \times l \times w$$

$$D^2 : D^2 - d^2$$

$$D^2 : D^2 - (\frac{1}{2}D)^2$$

$$D^2 : D^2 - \frac{1}{4}D^2$$

$$D^2 : \frac{3}{4}D^2$$

$$1 : \frac{3}{4} \quad \dots \dots \dots \quad \text{(ii)}$$

From (i) and (ii) it can be seen that by boring a concentric hole through a shaft, the diameter of the hole being half the shaft diameter, only one-sixteenth of the strength is lost but one-quarter of the weight is removed.

EXPERIMENT TO DETERMINE MODULUS OF RIGIDITY

One method of finding the modulus of rigidity of a material is to take a specimen in the form of a straight round bar, grip it at one end, apply various values of twisting moment at the other end and measure the angle of twist produced by each applied twisting moment. These values are plotted on a graph and the plotted points should lie exactly in a straight line because the angle of twist varies directly as the twisting moment. Due to slight errors of observation of measurement during the experiment, some points may not be in line, therefore the graph is drawn as an average between the high and low points in an attempt to obtain the true relationship. All other quantities in that part of the fundamental torsion equation connecting angle of twist and twisting moment being simple measurable dimensions which do not vary (length, diameter, etc.), the modulus of rigidity can be calculated from:

$$\frac{T}{J} = \frac{G\theta}{l} \quad G = \frac{Tl}{\theta J}$$

Example. The following observations were recorded during a torsion test on a steel bar 12 mm diameter and 1.8 m long. Plot a graph of twisting moment and angle of twist and find the modulus of rigidity of this steel.

Twisting moment (N m)	4	6	8	10	12
Angle of twist (deg.)	2.2	3.3	4.3	5.4	6.5

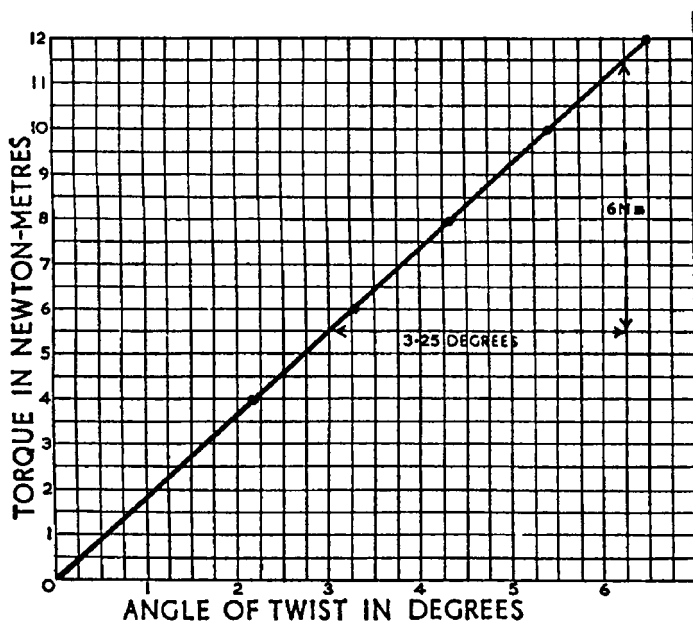


Fig. 196

Choosing two points on the graph such as shown in Fig. 196,

$$T = 6 \text{ N m}$$

$$\text{when } \theta = 3.25 \div 57.3 \text{ rad}$$

Inserting these values and the given dimensions of the bar into the expression,

$$\begin{aligned}
 G &= \frac{Tl}{\theta J} \\
 &= \frac{57.3 \times 32 \times 6 \times 1.8}{3.25 \times \pi \times 0.012^4} \\
 &= 9.354 \times 10^{10} \text{ N/m}^2 \\
 &= 93.54 \text{ GN/m}^2 \text{ or } 93.54 \text{ kN/mm}^2 \text{ Ans.}
 \end{aligned}$$

f TORSIONAL RESILIENCE

Referring to Fig. 197, let the torque be produced by a force F at a leverage of L to twist a shaft through θ rad. The angle of twist is proportional to the twisting moment applied, therefore the force is gradually applied from zero to the maximum of F while it moves through a linear distance of $L \times \theta$.

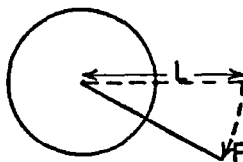


Fig. 197

$$\begin{aligned}\text{Work done} &= \text{Average force} \times \text{distance} \\ &= \frac{0 + F}{2} \times L \times \theta \\ &= \frac{FL\theta}{2}\end{aligned}$$

$F \times L$ is the final torque represented by T

$$\text{Work done to twist shaft} = \frac{T\theta}{2}$$

Provided the shaft is strained within the elastic limit, the work done to twist it will be stored up as elastic strain energy in the twisted shaft. This is termed *torsional resilience* and, as with direct resilience (see Chapter 9), can be expressed in terms of stress.

$$\begin{aligned}\text{Torsional resilience} &= \text{Work done to twist shaft} \\ &= \frac{T\theta}{2}\end{aligned}$$

$$\text{From } \frac{T}{J} = \frac{\tau}{r}$$

$$T = \frac{\pi D^3 \tau}{16} \text{ for a solid shaft}$$

$$\text{From } \frac{\tau}{r} = \frac{G\theta}{l}$$

$$\theta = \frac{\tau l}{rG} = \frac{2\tau l}{DG}$$

Substituting these values for T and θ ,

$$\begin{aligned}\text{Torsional resilience} &= \frac{\pi \times D^3 \times \tau \times 2 \times \tau \times l}{16 \times 2 \times D \times G} \\ &= \frac{\pi \times D^2 \times \tau^2 \times l}{16 \times G}\end{aligned}$$

Volume of shaft $= \frac{\pi}{4} D^2 \times l$, therefore these terms can be taken out and replaced by volume.

$$\text{Torsional resilience} = \frac{\tau^2 \times V}{4G}$$

RELATIONSHIP BETWEEN TORQUE AND POWER

Consider a steady force of F applied at a leverage of L turning a shaft,

$$\begin{aligned}\text{Work done to turn shaft one revolution} \\ &= \text{force} \times \text{distance} \\ &= F \times 2\pi L \quad \text{Nm i.e. } J\end{aligned}$$

If the shaft is running at n rev/s,

$$\begin{aligned}\text{Work done per second} &= \text{power} \\ &= F \times 2\pi L \times n \quad \text{J/s i.e. } W\end{aligned}$$

$$\text{But, } F \times L = \text{Torque}$$

$$\text{Power} = 2\pi T n$$

$$2\pi n \text{ is the angular velocity in rad/s} = \omega$$

$$\text{Power} = \text{Torque} \times \omega$$

$$P = T\omega$$

MAXIMUM AND MEAN TORQUE

It is important to note that the above expression involves the *mean* torque to transmit a given power. In a reciprocating engine the torque varies throughout a revolution, the variation depending upon the cycle of operations within the cylinder and the number and arrangement of the cranks. It is obvious that a shaft must be of sufficient strength to carry the *maximum* torque and the diameter must be calculated from this value, therefore the ratio of maximum to mean torque should be given when dealing with problems

involving reciprocating engines. In turbines the torque is steady and the mean torque is taken to be equal to the maximum.

The expressions previously derived can therefore be written:

$$\begin{aligned}\text{Maximum torque} &= \frac{\pi D^3 \tau}{16} \text{ for a solid shaft} \\ &= \frac{\pi (D^4 - d^4) \tau}{16D} \text{ for a hollow shaft}\end{aligned}$$

$$\text{Mean torque} = \frac{\text{power}}{\omega}$$

$$\text{i.e. } T = \frac{P}{\omega}$$

Example. A solid shaft from a turbine is to transmit 2000 kW at 3000 rev/min. Calculate the diameter of the shaft allowing a stress of 35 MN/m².

$$\frac{3000 \text{ rev/min} \times 2\pi}{60} = 100\pi \text{ rad/s}$$

$$\text{Torque} = \frac{\pi D^3 \tau}{16} = \frac{\text{power}}{\omega}$$

$$\frac{\pi \times D^3 \times 35 \times 10^6}{16} = \frac{2000 \times 10^3}{100 \times \pi}$$

$$\begin{aligned}D &= \sqrt[3]{\frac{2000 \times 10^3 \times 16}{\pi \times 35 \times 10^6 \times 100 \times \pi}} \\ &= 0.09747 \text{ m} = 97.47 \text{ mm} \quad \text{Ans.}\end{aligned}$$

Example. Calculate the maximum power that can be transmitted through a solid shaft 380 mm diameter when driven by a reciprocating engine at 110 rev/min if the stress is not to exceed 40 MN/m² and the ratio of maximum to mean torque is 1.4 to 1.

$$\frac{110 \text{ rev/min} \times 2\pi}{60} = 11.52 \text{ rad/s}$$

$$\text{Mean torque} = \frac{\text{power}}{\omega}$$

$$\text{Maximum torque} = 1.4 \times \text{mean torque}$$

$$= \frac{1.4 \times \text{power}}{11.52} \quad \dots \dots \dots (i)$$

$$\text{Also, maximum torque} = \frac{\pi D^3 \tau}{16}$$

$$= \frac{\pi \times 0.38^3 \times 40 \times 10^6}{16} \quad \dots \quad (ii)$$

$$\frac{1.4 \times \text{power}}{11.52} = \frac{\pi \times 0.38^3 \times 40 \times 10^6}{16}$$

$$\text{power} = \frac{11.52 \times \pi \times 0.38^3 \times 40 \times 10^6}{1.4 \times 16}$$

$$= 3.547 \times 10^6 \text{ W}$$

$$= 3.547 \text{ MW or } 3547 \text{ kW} \quad \text{Ans.}$$

TRANSMISSION OF POWER THROUGH COUPLING BOLTS

The bolts connecting shaft couplings carry the transmitted torque from one shaft to the next, shear stress is induced in them and the magnitude of this stress will depend upon the total sectional area of bolt material carrying the load and the radius of the circle on which the bolts are pitched.

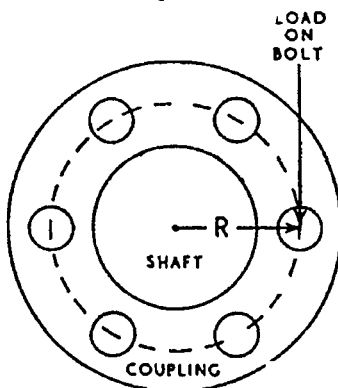


Fig. 198

Let shear stress in bolts $= \tau_B$

Cross-sect. area of each bolt $= \frac{\pi}{4} d^2$

Radius of pitch circle $= R$

Number of bolts in coupling $= N$

Torque transmitted by each bolt

$$= \text{Load on one bolt} \times \text{Radius of pitch circle.}$$

= Stress in bolt $\times$ Cross. sect. $\times$ Radius of pitch circle

$$= \tau_B \times \frac{\pi}{4} d^2 \times R$$

Total torque transmitted by all bolts in coupling

= Torque transmitted by one bolt $\times$ number of bolts

$$= \tau_B \times \frac{\pi}{4} d^2 \times R \times N$$

This must be equal to the torque transmitted by the shafting, therefore this can be equated to any one of the formulae for the torque in shafts previously given, depending upon the data given in the problem.

Example. The couplings of a 350 mm diameter solid shaft are 700 mm diameter and there are 8 bolts per coupling on a pitch circle diameter of 525 mm. Under maximum working conditions the stress in the shaft is 42 MN/m². Find the diameter of the coupling bolts allowing a stress in them of 35 MN/m².

Torque transmitted by bolts = Torque transmitted by shaft

$$\tau_B \times \frac{\pi}{4} d^2 \times R \times N = \frac{\pi}{16} D^3 \times \tau$$

$$35 \times 10^6 \times \frac{\pi d^2}{4} \times 0.2625 \times 8 = \frac{\pi}{16} \times 0.35^3 \times 42 \times 10^6$$

$$d = \sqrt{\frac{0.35^3 \times 42 \times 4}{35 \times 0.2625 \times 8 \times 16}}$$

$$= 0.07825 \text{ m} = 78.25 \text{ mm Ans.}$$

RECIPROCATING ENGINE MECHANISM

The force on the piston of a reciprocating engine is transmitted through the piston rod to the crosshead and sends a thrust through the connecting rod to the crank pin to exert a turning moment on the shaft.

The method of finding the thrust in the connecting rod for a given piston force and crank angle was explained in Chapter 1.

The connecting rod thrust multiplied by the perpendicular distance from its line of action to the shaft centre, is the turning moment applied to the shaft.

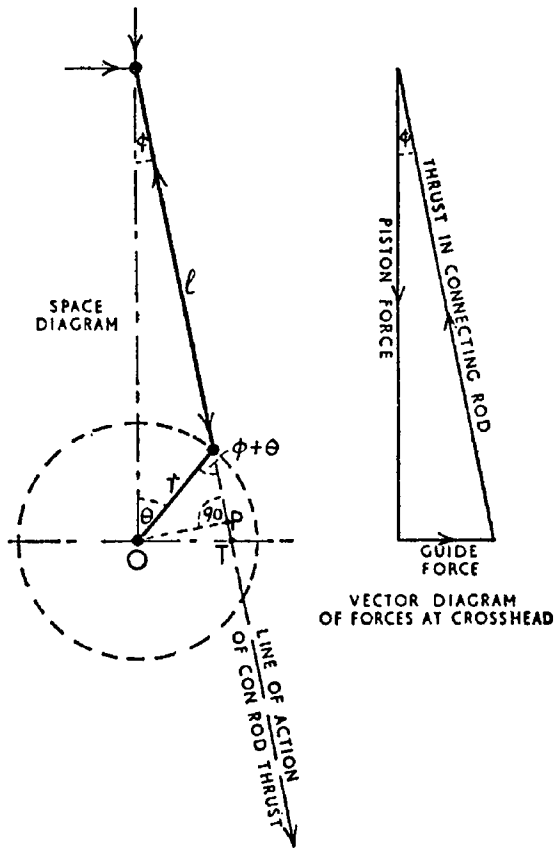


Fig. 199

Fig. 199 shows the crank at θ° past top centre. The perpendicular distance from the line of action of the connecting rod thrust to the shaft centre is marked OP , this is effective leverage, and:

$$\begin{aligned}\text{Turning moment} &= \text{Force} \times \text{perpendicular distance} \\ &= \text{thrust in conn. rod} \times OP\end{aligned}$$

Example. The effective pressure on the piston of a diesel engine is 17.25 bar ($= 17.25 \times 10^5 \text{ N/m}^2$) when the crank is 60° past top dead centre. The diameter of the piston is 500 mm , stroke 900 mm , and connecting rod length 1575 mm . Find the turning moment on the crank at this position.

$$\begin{aligned}
 \text{Piston force} &= \text{pressure} \times \text{area of piston} \\
 &= 17.25 \times 10^5 \times 0.7854 \times 0.5^2 \\
 &= 3.387 \times 10^5 \text{ N} = 338.7 \text{ kN}
 \end{aligned}$$

Referring to space diagram of Fig. 199,

$$\begin{aligned}
 \frac{l}{\sin \theta} &= \frac{r}{\sin \phi} \\
 \sin \phi &= \frac{0.45 \times \sin 60}{1.575} = 0.2474 \\
 \phi &= 14^\circ 19' \\
 OP &= r \times \sin (\phi + \theta) \\
 &= 0.45 \times \sin 74^\circ 19' = 0.4332 \text{ m}
 \end{aligned}$$

Referring to vector diagram of Fig. 199,

$$\begin{aligned}
 \text{Thrust in conn. rod} &= \frac{\text{Piston force}}{\cos \phi} \\
 &= \frac{338.7}{\cos 14^\circ 19'} = 349.5 \text{ kN} \\
 \text{Turning moment} &= \text{Thrust in conn. rod} \times OP \\
 &= 349.5 \times 0.4332 \\
 &= 151.5 \text{ kN m} \qquad \text{Ans.}
 \end{aligned}$$

ALTERNATIVE METHOD

Turning moment = Thrust in conn. rod $\times$ OP
and,

$$\text{Thrust in conn. rod} = \frac{\text{Piston force}}{\cos \phi}$$

Now, examining the space diagram of Fig. 199, note that angle POT is equal to ϕ , therefore,

$$OP = OT \times \cos \phi$$

Substituting,

$$\begin{aligned}
 \text{Turning moment} &= \text{Thrust in conn. rod} \times OP \\
 &= \frac{\text{Piston force}}{\cos \phi} \times OT \times \cos \phi \\
 &= \text{Piston force} \times OT
 \end{aligned}$$

The distance OT is the measurement from the shaft centre to the point where the line of action of the connecting rod thrust cuts the horizontal centre line as shown in Fig. 199 and this measurement can conveniently be found graphically by drawing the crank pin circle and connecting rod to scale.

This is particularly useful because one scale drawing of the crank pin circle can be used for measuring the distances OT for a series of different crank angles, these multiplied by the effective piston forces obtained from the indicator diagram at the points of the stroke relative to the crank angles, produces a series of values of torque which can be plotted as a graph to show the variations of torque throughout a cycle of operations.

CRANK EFFORT

The effects of the thrust of the connecting rod on the crank pin are (i) to turn the shaft, (ii) to exert either compression or tension in the crank webs. The values of (i) and (ii) are the rectangular components of the force in the connecting rod as shown in Fig. 200.

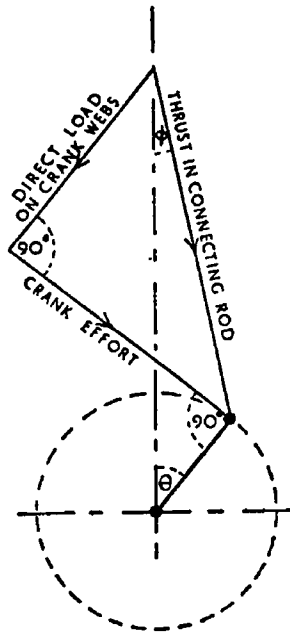


Fig. 200

One component, the force which effectively turns the shaft, is at right angles to the crank. The other component which causes compression or tension in the webs is in direct line with the crank.

The force which effectively turns the shaft is termed the *crank effort* and may be calculated from the force diagram, Fig. 200, alternatively, as the product of Crank Effort and Length of Crank is the Turning Moment, then,

$$\text{Crank Effort} = \frac{\text{turning moment}}{\text{crank length}}$$

HYDRAULIC STEERING GEAR

The arrangement of a hydraulic steering gear is based on the principle of Rapson's Slide. The force applied by the hydraulic ram on the swivel crosshead through which the tiller arm slides is the product of the hydraulic oil pressure and the cross-sectional area of the ram. Let this ram force be represented by F (Fig. 201).

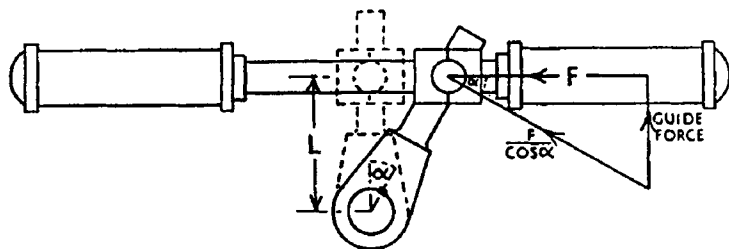


Fig. 201

The ram force F tends to cause the block to slide along the tiller arm towards the rudder stock and this is prevented by the supporting force of the guide acting at right angles to the ram force. The resultant of the ram force and the guide force is the effective force acting perpendicularly on the tiller causing it to rotate and turn the rudder stock.

If the tiller is at α° from mid-position, the resultant force is $F \div \cos \alpha$. If the distance from centre of rudder stock to the centre-line of the rams is L , the distance from rudder stock centre to crosshead is $L \div \cos \alpha$ and this is the perpendicular distance from the application of the effective force to the rudder stock centre.

$$\begin{aligned}\text{Turning moment} &= \frac{F}{\cos \alpha} \times \frac{L}{\cos \alpha} \\ &= \frac{FL}{\cos^2 \alpha}\end{aligned}$$

When α is zero, that is when the rudder is in mid-position, $\cos^2 \alpha = 1$ and then the turning moment is simply $F \times L$.

As the rudder moves from mid-position and α increases, the value of $\cos^2 \alpha$ decreases, therefore for a given constant working hydraulic pressure the turning moment increases as the rudder moves further from its mid-position. This is one important advantage of the hydraulic steering gear over others because, when the ship is moving ahead the turning moment required to move the rudder becomes greater as the angle of helm is increased.

Fig. 201 is a diagrammatic sketch of the arrangement of a two-ram steering gear in which one ram at a time exerts a turning effect on the rudder. In a four-ram system there is a similar pair on either side of the rudder stock and therefore two diagonally opposite rams act together in exerting the turning moment, this gives twice the turning effect as a two-ram steering gear (a couple).

Example. In a two-ram hydraulic steering gear, the rams are 250 mm diameter, the rudder stock is 400 mm diameter and the distance from centre of rudder stock to centre-line of rams is 800 mm. Calculate the stress in the rudder stock when the rudder is in mid-position and the hydraulic pressure is 70 bar.

$$\begin{aligned}\text{Ram force on tiller} &= 70 \times 10^5 \times 0.7854 \times 0.25^2 \\ &= 3.436 \times 10^5 \text{ N}\end{aligned}$$

$$\text{Turning moment} = 3.436 \times 10^5 \times 0.8 \text{ N m}$$

For a solid shaft,

$$\begin{aligned}T &= \frac{\pi}{16} D^3 \tau \\ \tau &= \frac{16 \times T}{\pi \times D^3} \\ &= \frac{16 \times 3.436 \times 10^5 \times 0.8}{\pi \times 0.4^3} \\ &= 2.187 \times 10^7 \text{ N/m}^2 \\ &= 21.87 \text{ MN/m}^2 \quad \text{Ans.}\end{aligned}$$

fExample. In a four-ram electric-hydraulic steering gear, the relief by-pass valves are set to lift when the oil pressure is 75 bar. The diameter of the rams is 300 mm, distance from centre of rudder stock to centre-line of each pair of rams is 760 mm and the

maximum angle of helm is 35° from mid-position. The tiller arms are parallel round section from the end to 560 mm from the rudder stock centre and the maximum bending stress in them may be taken as occurring at this section. Calculate (i) the diameter of the rudder stock to limit the maximum torsional stress to 75 MN/m^2 , and (ii) the diameter of the tiller arms to limit the maximum bending stress to 105 MN/m^2 .

$$\text{Max. force on ram} = 75 \times 10^5 \times 0.7854 \times 0.3^2 \text{ N}$$

Force applied at right angles to tiller

$$= \frac{75 \times 10^5 \times 0.7854 \times 0.3^2}{\cos 35^\circ}$$

$$= 6.471 \times 10^5 \text{ N}$$

Leverage from crosshead to rudder stock centre

$$= \frac{0.76}{\cos 35^\circ} = 0.9276 \text{ m}$$

Turning moment on rudder stock from two rams

$$= 6.471 \times 10^5 \times 0.9276 \times 2 \text{ N m}$$

For a solid round section, $T = \frac{\pi}{16} D^3 \tau$

$$\frac{\pi}{16} \times D^3 \times 75 \times 10^6 = 6.471 \times 10^5 \times 0.9276 \times 2$$

$$D = \sqrt[3]{\frac{16 \times 6.471 \times 0.9276 \times 2}{\pi \times 75 \times 10^6}}$$

$$= 0.4336 \text{ m} = 433.6 \text{ mm} \quad \text{Ans. (i)}$$

Length of tiller arm from crosshead to neck of parallel part which is 0.56 m from rudder stock centre

$$= 0.9276 - 0.56 = 0.3676 \text{ m}$$

Bending moment at this section

$$= 6.471 \times 10^5 \times 0.3676 \text{ N m}$$

For a round section subject to bending:

$$\sigma = \frac{32M}{\pi D^3}$$

$$D^3 = \frac{32 \times M}{\pi \times \sigma}$$

$$D = \sqrt[3]{\frac{32 \times 6.471 \times 10^5 \times 0.3676}{\pi \times 105 \times 10^6}}$$

$$= 0.2847 \text{ m} = 284.7 \text{ mm} \quad \text{Ans. (ii)}$$

f DEFLECTION OF A CLOSELY COILED
HELICAL SPRING

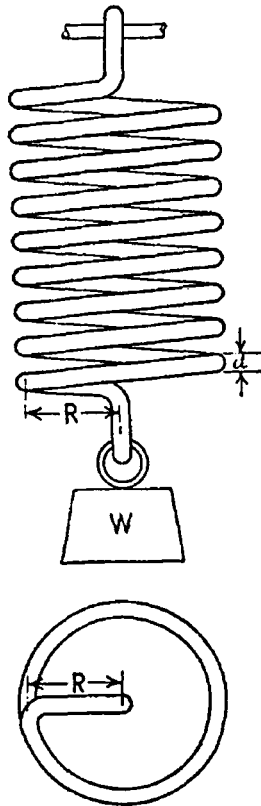


Fig. 202

If the coils of a helical spring are closely pitched, an axial load applied produces pure twist of the spring material, the twisting moment being the product of the load and the mean radius of the coils. It can therefore be considered in the same manner as the twisting of a straight shaft, the length of spring wire under twist being the length of each coil multiplied by the number of coils. The length of one coil can be taken simply as the mean circumference ($2\pi R$) because the spring is closely coiled (Fig. 202).

Let W = axial load,
 R = mean radius of coils,
 D = mean diameter of coils,
 d = diameter of coil wire,

the stress in the wire can be calculated thus:

$$T = \frac{\pi}{16} d^3 \tau$$

$$W \times R = \frac{\pi}{16} d^3 \tau$$

$$\tau = \frac{16WR}{\pi d^3} \text{ or } \frac{8WD}{\pi d^3}$$

The deflection (δ) is the axial stretch of the spring and an expression connecting axial load and axial deflection can be obtained by equating the external work applied to the work required to twist the material.

Let N = number of coils,

δ = axial deflection,

External work applied by W = Internal work to twist wire

Average force $\times$ distance = Average torque $\times$ angle of twist

$$\frac{1}{2} (O + W) \times \delta = \frac{1}{2} (O + T) \times \theta$$

$$W \times \delta = T \times \theta \quad \dots \dots \dots (i)$$

Finding suitable equivalents of T and θ for substitution:

$$\frac{T}{J} = \frac{G\theta}{l} \quad \theta = \frac{Tl}{JG}$$

$$T = W \times R$$

$$l = \text{length of wire} = 2\pi RN$$

$$J = \frac{\pi}{32} d^4$$

$$\theta = \frac{Tl}{JG}$$

$$= \frac{32 \times W \times R \times 2\pi RN}{\pi \times d^4 \times G}$$

Substituting for T and θ into (i)

$$W \times \delta = T \times \theta$$

$$W \times \delta = \frac{W \times R \times 32 \times W \times R \times 2\pi RN}{\pi \times d^4 \times G}$$

$$\delta = \frac{64WR^3N}{Gd^4} \text{ or } \frac{8WD^3N}{Gd^4}$$

f Example. A closely coiled helical spring is made of 6 mm diameter steel wire, the mean diameter of the coils is 60 mm and there are 8 coils. Taking the modulus of rigidity as 100 kN/mm² find the deflection and the stress in the wire when carrying an axial load of 240 N.

$$\delta = \frac{64WR^3N}{Gd^4}$$

Working in mm:

$$\begin{aligned}\delta &= \frac{64 \times 240 \times 30^3 \times 8}{100 \times 10^3 \times 6^4} \\ &= 25.6 \text{ mm} \quad \text{Ans. (i)}\end{aligned}$$

$$T = \frac{\pi}{16} d^3 \tau$$

$$W \times R = \frac{\pi}{16} d^3 \tau$$

$$\begin{aligned}\tau &= \frac{16 \times W \times R}{\pi \times d^3} \\ &= \frac{16 \times 240 \times 30}{\pi \times 6^3} \\ &= 169.7 \text{ N/mm}^2 \\ &= 169.7 \times 10^6 \text{ N/m}^2 \\ &= 169.7 \text{ MN/m}^2 \quad \text{Ans. (ii)}\end{aligned}$$

TEST EXAMPLES 11

1. A hollow shaft is 400 mm diameter outside and 250 mm diameter inside and transmits a torque of 480 kN m. Calculate the shear stress in the material and the angle of twist over a length of 7.5 m. Take $G = 92.5 \text{ GN/m}^2$.

2. A solid shaft is to be replaced by a hollow shaft. The hollow shaft is to be made from a higher quality steel so that the safe working stress allowed can be 20% higher and its outside diameter is to be equal to the diameter of the solid shaft so that the same bearings can be used. Calculate (i) the diameter of the bore of the hollow shaft in terms of the outside diameter, (ii) the percentage saving in weight assuming that the densities of the steels of both shafts are equal.

3. A solid steel shaft 350 mm diameter has a brass liner shrunk on it over its entire length, the thickness of the liner being 25 mm. Taking G for steel as 85 GN/m^2 and G for brass as 38.5 GN/m^2 , calculate the maximum shear stresses in the shaft and the liner when the total torque transmitted between them is 200 kN m.

4. The pinion shaft of a double reduction geared turbine is 140 mm diameter, runs at 3000 rev/min and the stress in it is 48 MN/m^2 . The main shaft is 445 mm diameter and runs at 90 rev/min. Assuming that the same power is transmitted by both shafts, find the stress in the main shaft.

5. A torsion-meter on a tunnel shaft 360 mm diameter registers an angle of twist of 0.3° over a length of 2.5 m when running at 115 rev/min. Taking the modulus of rigidity of the shaft material as 93 GN/m^2 , find the power transmitted.

6. A propeller shaft 380 mm diameter running at 90 rev/min drives a ship at 19 knots, the total resistance of the ship through the water at this speed is 320 kN. Taking the propeller efficiency as 0.75, find (i) the power transmitted by the shaft, (ii) the twisting moment in the shaft, (iii) the torsional stress in the shaft. 1 knot = 1.852 km/h.

7. An engine develops 4500 kW at 105 rev/min and the ratio of maximum to mean torque is 1.25 to 1.

(a) Calculate the diameter of the main shaft coupled direct to the engine to transmit this power allowing a stress of 55 MN/m^2 and assuming the shaft to be (i) solid, (ii) hollow with a bore equal to half the outside diameter.

(b) Calculate the percentage saving in weight by fitting the hollow shaft as compared with the solid shaft.

8. Find the diameter of the coupling bolts to connect shafts which are to transmit 3000 kW when running at 100 rev/min if 8 bolts are to be fitted on a pitch circle diameter of 610 mm and allowing a stress of 38 MN/m² in the bolts.

9. The stroke of an internal combustion engine is 630 mm and the connecting rod length is 1200 mm. When the crank is 35° past top centre the load on the piston is 250 kN. Find the turning moment at this point.

10. The rams of a four-ram hydraulic steering gear are 250 mm diameter, the rudder stock is 430 mm diameter and the distance from rudder stock centre to centre-line of rams is 760 mm. Calculate the stress in the rudder stock at the instant the rudder is moving past mid-position and the hydraulic pressure on the rams is 70 bar (= 70 × 10⁵ N/m²).

f11. Prove that the elastic strain energy in a hollow shaft subjected to a torsional stress of τ is given by the expression:

$$\frac{\tau^2}{4G} \times \frac{D^2 + d^2}{D^2} \times \text{Volume of shaft.}$$

f12. The drive to a centrifuge consists of a solid steel shaft 30 mm in diameter which is a sliding fit in a hollow shaft of outside diameter 38 mm. The two shafts are secured by a shear pin fitted across the diameter of both shafts perpendicular to the shafts axis. The shear stress in the solid shaft is not to exceed 20 MN/m². Calculate:

- (a) the maximum torque that can be transmitted,
- (b) The maximum shear stress in a hollow shaft when transmitting this maximum torque,
- (c) a suitable diameter for the pin (which will shear at a stress of 80 MN/m²) to protect the solid shaft from excessive stress.

f13 In a two-ram hydraulic steering gear the diameter of the rams is 280 mm, diameter of rudder stock 350 mm, distance from rudder stock centre to centre-line of rams 860 mm and the maximum angle of helm is 35°. Find the pressure at which the by-pass relief valves should be set to lift in order to limit the maximum shear stress in the rudder stock to 77 MN/m².

f14. In a close coiled helical spring there are 27 coils of solid round wire 3 mm diameter, the mean diameter of the coils being 30 mm. If the modulus of rigidity of the material is 93 kN/mm² find the deflection when carrying an axial load of 50 N. Derive any formula used in the solution.

f15. A close coiled helical spring consists of 24 coils of tubular steel, the mean diameter of the coils is 75 mm, the outside diameter of the tube is 12.5 mm and its thickness is 2.5 mm. If the stress in the material is not to exceed 70 N/mm^2 find the maximum axial load the spring can carry and the deflection under this load. Take $G = 90 \text{ kN.mm}^2$.